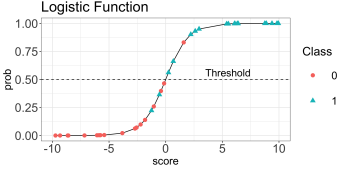
# Introduction to Machine Learning

# Supervised Classification In a Nutshell

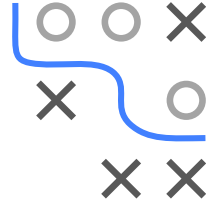



## Learning goals

- Understand basic concept of classifiers
- Know concepts of probabilistic and scoring classifier
- Know distinction between discriminant and generative approach
- Understand ideas of logistic regression and Naive Bayes

# CLASSIFICATION TASKS

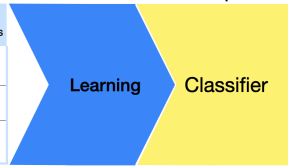
- Learn function that assigns categorical class labels to observations
- Each observation belongs to exactly one class
- The task can contain two (binary) or multiple (multi-class) classes



Training

Input: Labeled data

| Sex | Age | Health | Degree of Graduation | Level of Happiness |
|---|---|---|---|---|
| Female | 23 | Good | Bachelor | Not too happy |
| Male | 70 | Fair | High School | Pretty happy |
| Female | 48 | Excellent | High School | Very happy |

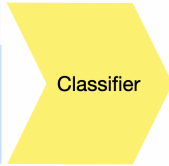



Output

Prediction

Input: Unlabeled data

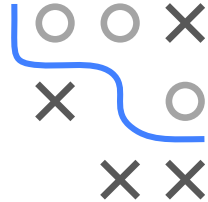
| Sex | Age | Health | Degree of Graduation | Level of Happiness |
|---|---|---|---|---|
| Male | 41 | Fair | Bachelor | ? |
| Male | 35 | Good | Bachelor | ? |
| Female | 22 | Fair | High School | ? |



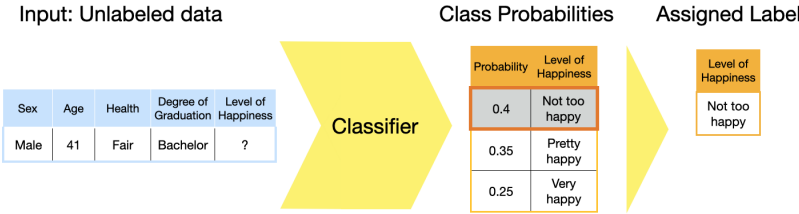

Prediction

| Level of Happiness |
|---|
| Not too happy |
| Pretty happy |
| Not too happy |

# BASIC DEFINITIONS



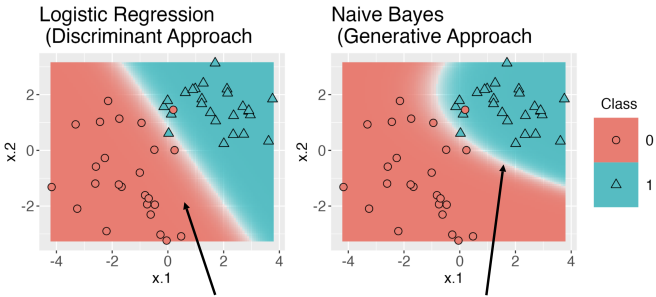
- For every observation a model outputs the probability (probabilistic classifier) or score (scoring classifier) of each class
- In the multi-class case, the class label is usually assigned by choosing the class with the maximum score or probability
- In the binary case, a class label is assigned by choosing the class whose probability or score exceeds a threshold value c



# BASIC DEFINITIONS / 2

Two fundamental approaches exist to construct a classifier:

- **Discriminant approach** asks “What is the best prediction for the class given these data?” (uses loss functions and empirical risk minimization)
- **Generative approach** asks “Which class tends to have data like these?” (models the feature distributions in each class separately)

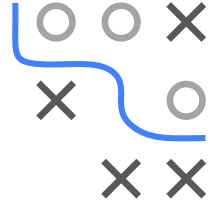


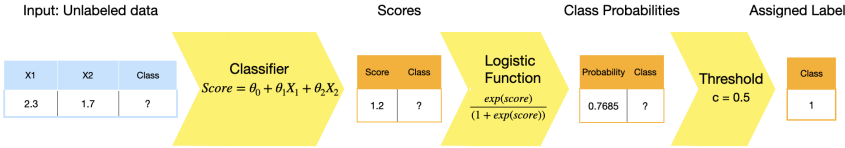

**Decision Region:** Region where all observations are assigned to the same class.

**Decision Boundary:** Points where all classes have the same probability/score.

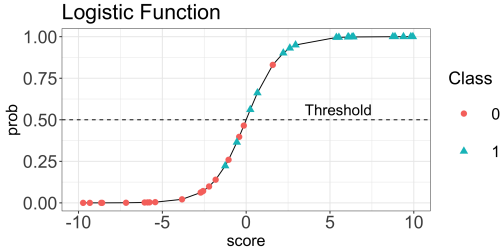
# LOGISTIC REGRESSION

- Logistic regression is a **discriminant approach** for binary classification. It turns scores into probabilities with the logistic function.
- We just need to compute the probability for **one** class (usually class 1).
- If the probability exceeds a threshold value **c** $\Rightarrow$ class 1 is predicted.



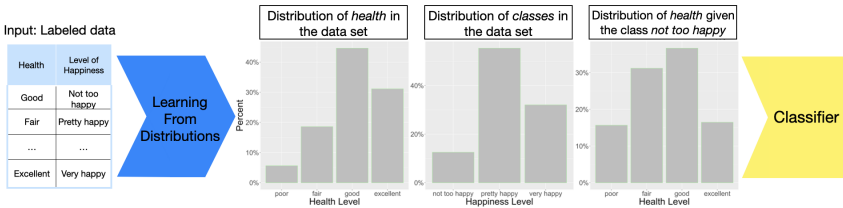


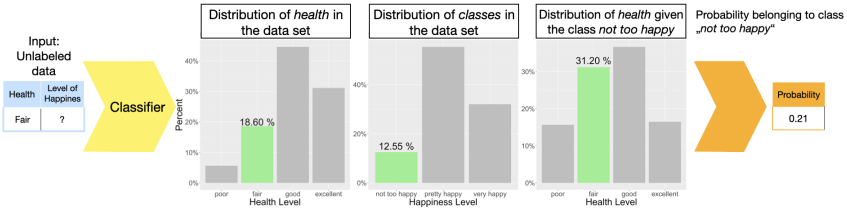
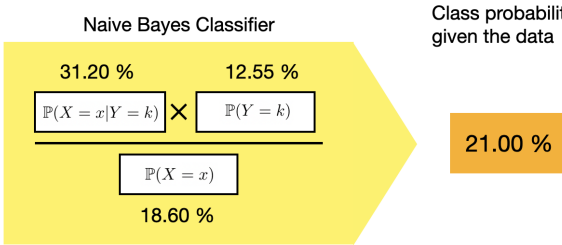

The logistic function puts all scores in order along an s-shaped line.



# NAIVE BAYES

- Naive Bayes is a **generative multi-class approach**. It computes the class probability for each class based on the training data.
- It considers the data distribution on three different levels:
  - Marginal distributions $\mathbb{P}(X)$ of each feature (in the entire data set)
  - Marginal distribution $\mathbb{P}(Y)$ of classes (in the entire data set)
  - Conditional distributions $\mathbb{P}(X|Y)$ of each feature in each class



# NAIVE BAYES / 2

- Example: Class probability of “not too happy” given health = “fair”: